



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to V.V.D. PETRO

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

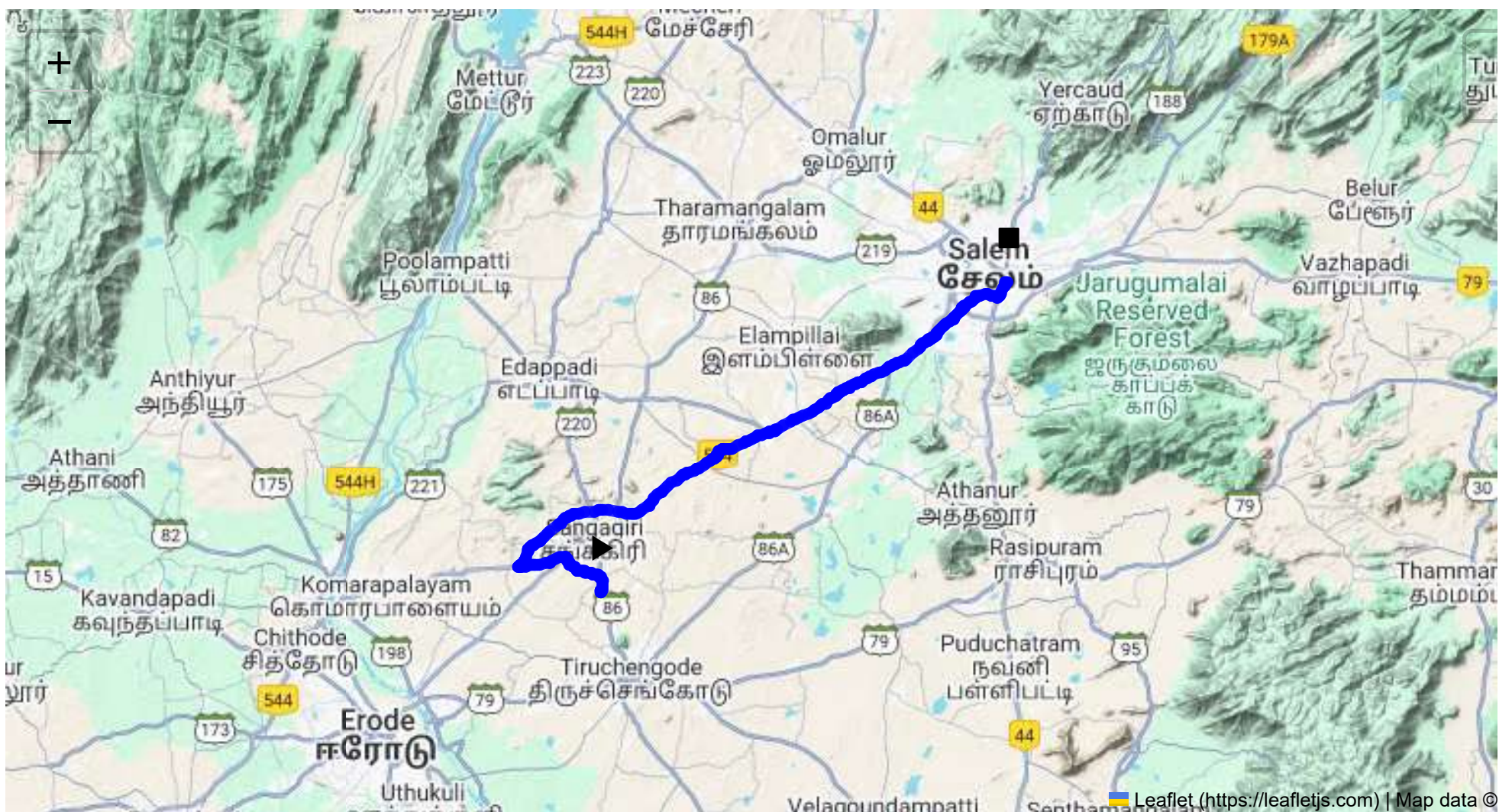
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 53.75 km**  
**Estimated Duration: 1.1 hours**  
**Adjusted Duration (Heavy Vehicle): 1.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.650399, 78.158669)**



## Welcome to the Journey Risk Management Study

### Route Overview:

The route covers approximately 53.75 kilometers, starting from Sangagiri and ending in Salem, Tamil Nadu. It involves several critical waypoints, mainly major roads and highways, such as the Sankari RS - Erode Privu Road, Bhavani Main Road, and NH544. Being a primary route, it's likely to intersect with other major roads and could include both urban and semi-urban areas.



## Typical Weather Conditions and Hazards:

- The region typically experiences a tropical climate, with hot summers and moderate rainfall during the monsoon season.
- Weather hazards include heavy rains causing flooding and reduced visibility, particularly between June and September.
- During summer (March to June), extremely high temperatures can lead to road surface damage and tire issues.

## Traffic Patterns:

- Traffic congestion could be significant in urban sections, especially near intersections like NH544.
- Peak hours are usually morning (8 AM – 10 AM) and evening (5 PM – 8 PM) on weekdays.
- Markets and school zones along the route further increase congestion during specific time slots.

## Road Quality and Infrastructure:

- NH544 is generally well-maintained, but smaller roads and certain stretches through urban areas may suffer from patchy maintenance.
- Potholes, uneven surfaces, and poorly marked intersections might pose challenges.

## Alternative Routes:

- In case of emergencies, taking a detour via nearby highways or main roads like NH79 could be considered, although they might increase travel time.
- Local roads in the vicinity of these main highways may provide short bypass options but could have very limited capacity for heavy vehicles.

## Local Regulations:

- Regulations on transporting hazardous materials include strict adherence to loading limits, proper vehicle markings, and obtaining necessary permits.
- Night-time restrictions might apply for transportation through densely populated areas.

## Historical Incidents:

- Data may indicate sporadic incidents involving heavy vehicles, often due to speeding, brake failure, or poor road conditions, especially on hilly or sharp-curved sections.

## Environmental Considerations:

- Sensitive areas include agricultural zones and residential sectors where care must be taken to prevent contamination.
- Waste management and spill response preparedness are critical in these regions.

## Communication Coverage:

- Major towns and city areas typically have good communication coverage, but rural stretches and minor roads may have spotty service.
- Satellite communication tools could be advisable as a backup.

## Emergency Response Times:

- Proximity to major towns like Erode and Salem shortens response times to an estimated 20-30 minutes in urban areas.
- Rural stretches or during adverse weather conditions could extend response times up to an hour.

## Overall Risk Assessment:

- This route is moderately risky due to traffic congestion, variable road quality, and weather inconsistencies.
- Careful planning for timing, adherence to regulations, and preparation for adverse weather conditions can mitigate many of these risks.

## Recommendations for Truck Drivers:

- Conduct regular vehicle checks, especially before embarking on routes with extreme temperature forecasts or post-rainy season.
- Stay updated on local traffic news and be prepared for detours during peak congestion times or in case of roadblocks.
- Have proper emergency kits that include communication devices, first aid, and tools for minor repairs.
- Additional training on recognizing and navigating through poorly maintained road sections and handling emergency situations might be beneficial.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.63266, 78.12699 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 3 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4 | Turn       | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 5 | Turn       | Medium     | 11.45352, 77.85698 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 6  | Turn      | Medium     | 11.46048, 77.85036 | 30 KM/Hr    |
| 7  | Turn      | Medium     | 11.46303, 77.84945 | 30 KM/Hr    |
| 8  | Turn      | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 9  | Turn      | High       | 11.45542, 77.81725 | 15 KM/Hr    |
| 10 | Turn      | High       | 11.45631, 77.81668 | 15 KM/Hr    |
| 11 | Turn      | High       | 11.61233, 78.10397 | 15 KM/Hr    |
| 12 | Turn      | Medium     | 11.61216, 78.10423 | 30 KM/Hr    |
| 13 | Turn      | Medium     | 11.64132, 78.15294 | 30 KM/Hr    |
| 14 | Turn      | High       | 11.64134, 78.15304 | 15 KM/Hr    |

### Emergency Locations

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 2  | hospital | C. N. Hospital                                                  | 11.55724, 78.0096         | 30 km/h     | Medium     |
| 3  | hospital | Annapoorna Medical College Hospital                             | 11.57693,<br>78.037627    | 30 km/h     | Medium     |
| 4  | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 5  | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 6  | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 9  | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 13 | hospital | Thayammal Hospitals                                             | 11.6237472,<br>78.1177631 | 30 km/h     | Medium     |
| 14 | hospital | Vanitha Hospital                                                | 11.6239931,<br>78.1184622 | 30 km/h     | Medium     |
| 16 | hospital | Universal Cancer Hospital                                       | 11.6336739,<br>78.1267767 | 30 km/h     | Medium     |
| 17 | hospital | Priyam Speciality Hospital                                      | 11.63446,<br>78.124388    | 30 km/h     | Medium     |
| 18 | clinic   | Salem Care Foundation                                           | 11.6371946,<br>78.1313713 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 19 | hospital | Aravind Eye Hospital               | 11.641927,<br>78.1392347  | 30 km/h     | Medium     |
| 20 | police   | Annadanapatti B 4 Police Station   | 11.6424031,<br>78.1404779 | 30 km/h     | Medium     |
| 21 | clinic   | Shri Bharthi Nursing Home          | 11.6420069,<br>78.1424099 | 30 km/h     | Medium     |
| 22 | police   | Police Control Room Police Station | 11.6425701,<br>78.1459515 | 30 km/h     | Medium     |
| 23 | hospital | Sree Vasantham Hospital            | 11.6420424,<br>78.1463903 | 30 km/h     | Medium     |
| 25 | hospital | Anbu Hospital                      | 11.6402823,<br>78.152387  | 30 km/h     | Medium     |
| 26 | clinic   | Shanti Nursing Home, Salem         | 11.6407328,<br>78.1532635 | 30 km/h     | Medium     |
| 27 | clinic   | Soundararajan Nursing Home         | 11.6379994,<br>78.1517239 | 30 km/h     | Medium     |
| 28 | clinic   | Ramadas Nursing Home               | 11.6449413,<br>78.1545496 | 30 km/h     | Medium     |
| 29 | hospital | G.V.K. Hospital                    | 11.6445919,<br>78.1555742 | 30 km/h     | Medium     |
| 30 | hospital | Life Care Hospital                 | 11.6452364,<br>78.1546993 | 30 km/h     | Medium     |
| 31 | clinic   | Sri Balaji Clinic                  | 11.645189,<br>78.154708   | 30 km/h     | Medium     |
| 32 | hospital | Janani Sree Eye Hospital           | 11.6445844,<br>78.1543916 | 30 km/h     | Medium     |
| 36 | clinic   | Ravi Eye Clinic                    | 11.646684,<br>78.1547669  | 30 km/h     | Medium     |
| 37 | hospital | Vasan Eye Care Hospital, Gugai     | 11.6471454,<br>78.1559283 | 30 km/h     | Medium     |
| 38 | hospital | Pp Hospital                        | 11.6474921,<br>78.1561723 | 30 km/h     | Medium     |
| 39 | hospital | Corporation Maternity Hospital     | 11.6524437,<br>78.154675  | 30 km/h     | Medium     |

## Crowded Spots

|    | type       | name                                                       | coordinates               | speed_limit | risk_level |
|----|------------|------------------------------------------------------------|---------------------------|-------------|------------|
| 0  | school     | KRP Matric. Hr. Sec School                                 | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 7  | university | Vinayaka Missions Vinayaka Missions<br>Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 8  | school     | Vinayaka Missions Annapoorani Matric<br>School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 10 | school     | DIET - Salem                                               | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 11 | school     | Dist Education Training                                    | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 12 | school     | Sri Vidhya Bharati Matric HR Sec School                    | 11.6186069,<br>78.1103239 | 30 km/h     | Medium     |
| 15 | school     | Sowdeshwari College                                        | 11.6330643,<br>78.1276021 | 30 km/h     | Medium     |
| 24 | school     | The gugai hr.sec school                                    | 11.6411104,<br>78.1472518 | 30 km/h     | Medium     |
| 33 | school     | Veeralashmi Vidhyalaya Higher<br>Secondary School          | 11.6414737,<br>78.1565634 | 30 km/h     | Medium     |
| 34 | school     | Jayarani matric school                                     | 11.6463187,<br>78.1527477 | 30 km/h     | Medium     |
| 35 | school     | Veeralakshmi Vidhyalaya Higher<br>Secondary School         | 11.6422707,<br>78.1567216 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High



**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.63266, 78.12699**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**  
**Risk Level: Medium**  
**Speed Limit: 30 KM/Hr**  
**Coordinates: 11.45357, 77.85789**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46048, 77.85036





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.46303, 77.84945**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.46318, 77.84913**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.45542, 77.81725**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45631, 77.81668

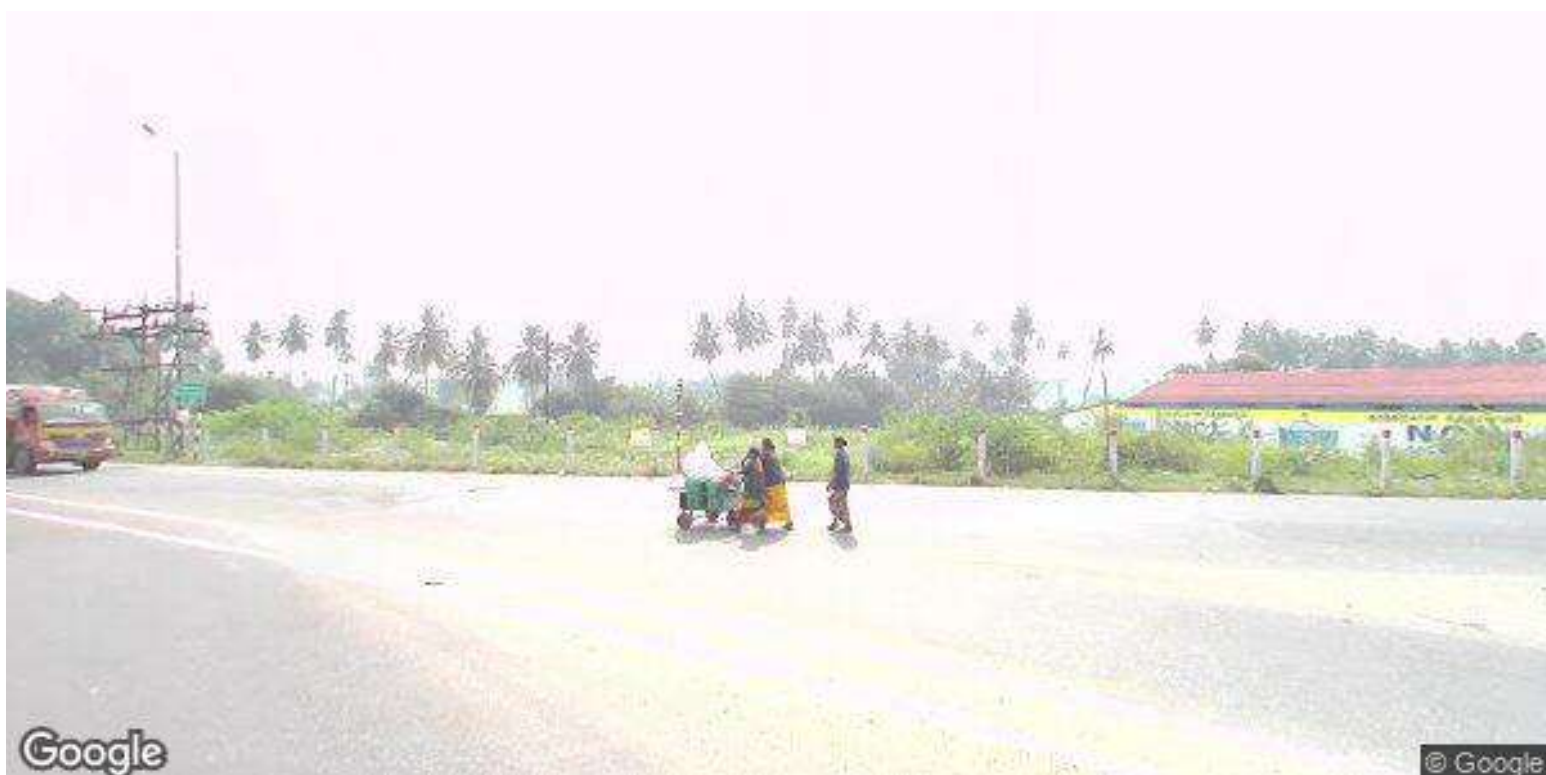


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.61233, 78.10397





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.61216, 78.10423



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64132, 78.15294



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.64134, 78.15304



